AAHLS (Lab 2)

1. 簡介

本報告探討 AAHLS（Lab2），內容分為兩個部分。第一部分使用 AXI-M 傳輸數據來實現 FIR，第二部分則改為 AXI-Stream 傳輸數據來實現 FIR。此外，其他介面皆會透過 AXI-Lite 來實現。本次 LAB 使用 KV260(xck26-sfvc784-2LV-c)進行實作，詳細內容會在後面呈現。

2. 報告內容

#1:

(1) 內容

本次 Lab2 是實作 11taps 的 FIR，其中#1 使用 AXI-Master 作為資料讀寫的 interface，具體硬體架構是使用一個 shift register 將 input data 接到 shift register 中，透過 tap parameters 和 shift register 內部的 data 做 convolution 以實現 FIR。由於此 Lab 著重在 IO 的設定，硬體優化部分，包括 PIPELINE、UNROLL、ARRAY PARTITION 放在 QUESTION 進行回答。這次 HLS 主要有兩個 LOOP，分別是 SHIFT_ACC_LOOP 以及 XFER_LOOP。SHIFT_ACC_LOOP 主要是進行輸出，每完成一次 LOOP 就會輸出一個對應的值，而 XFER_LOOP 只要控制所有輸入 data，完成 XFER_LOOP 代表對所有輸入 data 完成運算。

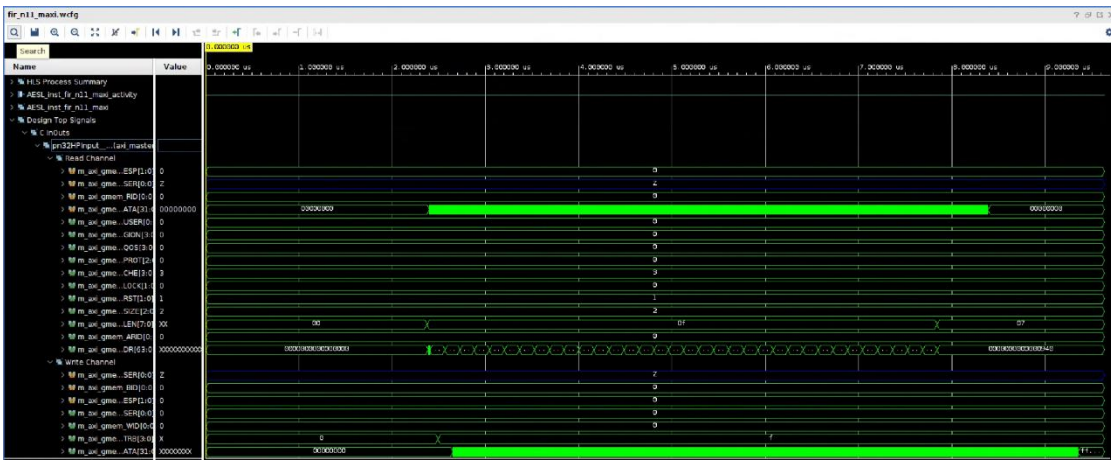
#1 使用的是 AXI-Master 進行資料讀寫，因此需要啟用 Zynq MPSoC 的 Slave HP (High-Performance) Port，使 Kernel 能夠存取 DDR 進行讀寫。這需要在 Block Diagram 中手動啟用 Slave HP Port，完成相關設定後，即可產生 .hwh 和 .bit 檔案，供 Host 端操作。其中，.bit 是 bitstream 檔案，負責透過 fpga_manager 配置 fpga，而 hwh 則包含 mmio、ip 等訊息，方便 python 進行調用 kernel。

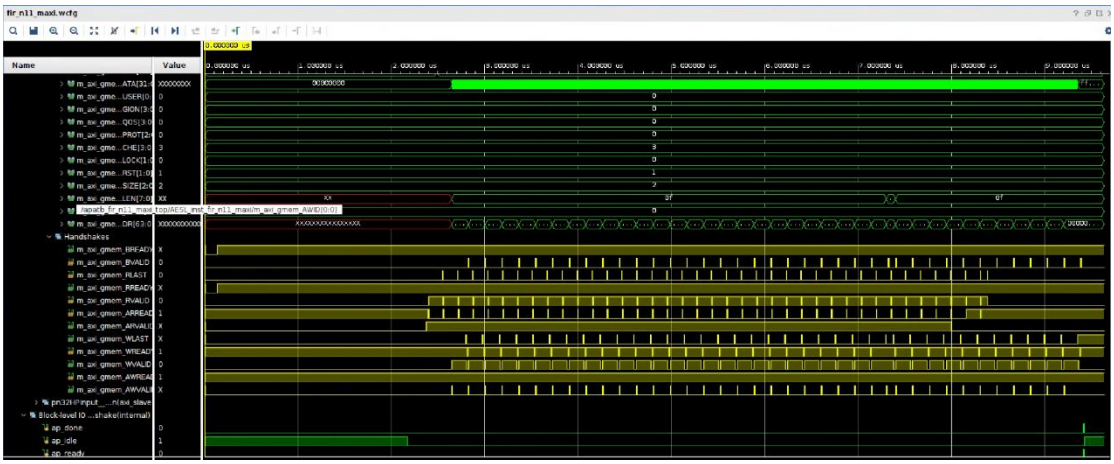
在 python code 的部分，和 Lab1 不同的是，此次 Lab 會先將 data 透過 allocate 存入 main memory 中，接著再進行 kernel 的 configuration，在配置過程中也會計算 DC gain 以方便後續進行 normalized。再來會將 main memory 的 data 透過 AXI-Master 傳輸到 kernel，然後開始計算後再將結果一樣透過 AXI-Master 傳回 main memory 並畫出結果。

(2) 相關截圖

```
================================================================
== Vitis HLS Report for 'fir_n11_maxi'
================================================================
* Date:           Fri Feb 28 02:04:01 2025

* Version:        2022.1 (Build 3526262 on Mon Apr 18 15:47:01 MDT 2022)
* Project:        hls_FIRNN11MAXI
* Solution:       solution1 (Vivado IP Flow Target)
* Product family: zynquplus
* Target device:  xck26-sfvc784-2LV-c


================================================================
== Performance Estimates
================================================================
+ Timing:
    * Summary:
    +--------+---------+----------+------------+
    |  Clock |  Target | Estimated| Uncertainty|
    +--------+---------+----------+------------+
    |ap_clk  |  10.00 ns|  7.300 ns|     2.70 ns|
    +--------+---------+----------+------------+

+ Latency:
    * Summary:
    +---------+---------+----------+----------+-----+-----+---------+
    |  Latency (cycles) |  Latency (absolute) |  Interval | Pipeline|
    |   min   |   max   |    min   |    max   | min | max |   Type  |
    +---------+---------+----------+----------+-----+-----+---------+
    |        ?|        ?|         ?|         ?|    ?|    ?|       no|
    +---------+---------+----------+----------+-----+-----+---------+

    + Detail:
        * Instance:
        +-----------------------------------------+-----------------------------+---------+---------+----------+----------+-----+-----+---------+
        |                                         |                             |  Latency (cycles) |  Latency (absolute) |  Interval | Pipeline|
        |                Instance                 |             Module          |   min   |   max   |    min   |    max   | min | max |   Type  |
        +-----------------------------------------+-----------------------------+---------+---------+----------+----------+-----+-----+---------+
        |grp_fir_n11_maxi_Pipeline_XFER_LOOP_fu_242 |fir_n11_maxi_Pipeline_XFER_LOOP |        ?|        ?|         ?|         ?|    ?|    ?|       no|
        +-----------------------------------------+-----------------------------+---------+---------+----------+----------+-----+-----+---------+

        * Loop:
        N/A


================================================================
== Utilization Estimates
================================================================
* Summary:
+-----------------+---------+-------+--------+--------+-----+
|      Name       | BRAM_18K|  DSP  |   FF   |   LUT  | URAM|
+-----------------+---------+-------+--------+--------+-----+
|DSP              |        -|      -|       -|       -|    -|
|Expression       |        -|      -|       0|      40|    -|
|FIFO             |        -|      -|       -|       -|    -|
|Instance         |        0|     33|    1467|    2466|    0|
|Memory           |        -|      -|       -|       -|    -|
|Multiplexer      |        -|      -|       -|     175|    -|
|Register         |        -|      -|     650|       -|    -|
+-----------------+---------+-------+--------+--------+-----+
|Total            |        0|     33|    2117|    2681|    0|
+-----------------+---------+-------+--------+--------+-----+
|Available        |      288|   1248|  234240|  117120|   64|
+-----------------+---------+-------+--------+--------+-----+
|Utilization (%)  |        0|      2|      ~0|       2|    0|
+-----------------+---------+-------+--------+--------+-----+
```

```
================================================================
== Synthesis Summary Report of 'fir_n11_maxi'
================================================================
+ General Information:
    * Date:           Fri Feb 28 02:04:02 2025
    * Version:        2022.1 (Build 3526262 on Mon Apr 18 15:47:01 MDT 2022)
    * Project:        hls_FIRNN11MAXI
    * Solution:       solution1 (Vivado IP Flow Target)
    * Product family: zynquplus
    * Target device:  xck26-sfvc784-2LV-c


+ Performance & Resource Estimates:

    PS: '+' for module; 'o' for loop; '*' for dataflow
    +-----------------------------------+-----+-------+---------+---------+----------+---------+-----+----------+-----+---------+-----------+-----------+-----+
    |              Modules              | Issue|       | Latency | Latency| Iteration|         | Trip |          |     |         |           |           |     |
    |              & Loops              | Type | Slack| (cycles)|  (ns)  |  Latency | Interval| Count| Pipelined| BRAM |   DSP   |     FF    |    LUT    | URAM|
    +-----------------------------------+-----+-------+---------+---------+----------+---------+-----+----------+-----+---------+-----------+-----------+-----+
    |+ fir_n11_maxi                     |     -|  0.00|        -|        -|         -|        -|    -|        no|    -| 33 (2%)| 2117 (~0%)| 2681 (2%)|    -|
    | + fir_n11_maxi_Pipeline_XFER_LOOP |     -|  0.00|        -|        -|         -|        -|    -|        no|    -| 33 (2%)|  463 (~0%)| 756 (~0%)|    -|
    |  o XFER_LOOP                      |     -|  7.30|        -|        -|         4|        1|    -|       yes|    -|        -|          -|          -|    -|
    +-----------------------------------+-----+-------+---------+---------+----------+---------+-----+----------+-----+---------+-----------+-----------+-----+


================================================================
== HW Interfaces
================================================================
* M_AXI
+-----------+------------+---------------+---------+--------+---------+---------+-------------+-------------+-----------+-----------+
| Interface | Data Width | Address Width | Latency | Offset | Register | Max Widen | Max Read    | Max Write   | Num Read  | Num Write |
|           | (SW->HW)   |               |         |        |          | Bitwidth  | Burst Length | Burst Length | Outstanding | Outstanding |
+-----------+------------+---------------+---------+--------+---------+---------+-------------+-------------+-----------+-----------+
| m_axi_gmem | 32 -> 32   | 64            | 0       | slave  | 0       | 0         | 16          | 16          | 16        | 16        |
+-----------+------------+---------------+---------+--------+---------+---------+-------------+-------------+-----------+-----------+

* S_AXILITE Interfaces
+---------------+------------+---------------+--------+----------+
| Interface     | Data Width | Address Width | Offset | Register |
+---------------+------------+---------------+--------+----------+
| s_axi_control | 32         | 7             | 16     | 0        |
+---------------+------------+---------------+--------+----------+

* S_AXILITE Registers
+---------------+----------------+--------+-------+--------+-----------------------------+-----------------------------------------------------------+
| Interface     | Register       | Offset | Width | Access | Description                 | Bit Fields                                                |
+---------------+----------------+--------+-------+--------+-----------------------------+-----------------------------------------------------------+
| s_axi_control | CTRL           | 0x00   | 32    | RW     | Control signals             | 0=AP_START 1=AP_DONE 2=AP_IDLE 3=AP_READY 7=AUTO_RESTART 9=INTERRUPT |
| s_axi_control | GIER           | 0x04   | 32    | RW     | Global Interrupt Enable Register | 0=Enable                                                  |
| s_axi_control | IP_IER         | 0x08   | 32    | RW     | IP Interrupt Enable Register | 0=CHAN0_INT_EN 1=CHAN1_INT_EN                             |
| s_axi_control | IP_ISR         | 0x0c   | 32    | RW     | IP Interrupt Status Register | 0=CHAN0_INT_ST 1=CHAN1_INT_ST                             |
| s_axi_control | pn32HPInput_1  | 0x10   | 32    | W      | Data signal of pn32HPInput  |                                                           |
| s_axi_control | pn32HPInput_2  | 0x14   | 32    | W      | Data signal of pn32HPInput  |                                                           |
| s_axi_control | pn32HPOutput_1 | 0x1c   | 32    | W      | Data signal of pn32HPOutput |                                                           |
| s_axi_control | pn32HPOutput_2 | 0x20   | 32    | W      | Data signal of pn32HPOutput |                                                           |
| s_axi_control | regXferLeng    | 0x28   | 32    | W      | Data signal of regXferLeng  |                                                           |
+---------------+----------------+--------+-------+--------+-----------------------------+-----------------------------------------------------------+

* TOP LEVEL CONTROL
+-----------+-----------+-----------+
| Interface | Type      | Ports     |
+-----------+-----------+-----------+
| ap_clk    | clock     | ap_clk    |
| ap_rst_n  | reset     | ap_rst_n  |
| interrupt | interrupt | interrupt |
| ap_ctrl   | ap_ctrl_hs |           |
+-----------+-----------+-----------+
```



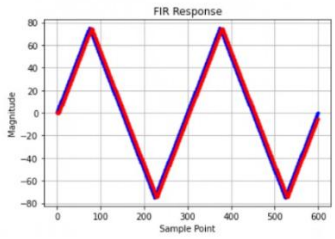


In [6]:

```python
# coding: utf-8

# In[ ]:


from __future__ import print_function

import sys, os
import numpy as np
from time import time
import matplotlib.pyplot as plt

sys.path.append('/home/xilinx')
os.environ['XILINX_XRT'] = '/usr'
from pynq import Overlay
from pynq import allocate

if __name__ == "__main__":
    print("Entry:", sys.argv[0])
    print("System argument(s):", len(sys.argv))

    print("Start of \"" + sys.argv[0] + "\"")

    ol = Overlay("/home/root/jupyter_notebooks/FIRN11MAXI.bit")
    ipFIRN11 = ol.fir_n11_maxi_0

    fiSamples = open("samples_triangular_wave.txt", "r+")
    numSamples = 0
    line = fiSamples.readline()
    while line:
        numSamples = numSamples + 1
        line = fiSamples.readline()

    inBuffer0 = allocate(shape=(numSamples,), dtype=np.int32)
    outBuffer0 = allocate(shape=(numSamples,), dtype=np.int32)
    fiSamples.seek(0)
    for i in range(numSamples):
        line = fiSamples.readline()
        inBuffer0[i] = int(line)
    fiSamples.close()

    numTaps = 11
    n32Taps = [0, -10, -9, 23, 56, 63, 56, 23, -9, -10, 0]
    #n32Taps = [1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1]
    n32DCGain = 0
    timeKernelStart = time()
    for i in range(numTaps):
        n32DCGain = n32DCGain + n32Taps[i]
        ipFIRN11.write(0x40 + i * 4, n32Taps[i])
    if n32DCGain < 0:
```

```python
    if n32DCGain < 0:
        n32DCGain = 0 - n32DCGain
    ipFIRN11.write(0x28, len(inBuffer0) * 4)
    ipFIRN11.write(0x10, inBuffer0.device_address)
    ipFIRN11.write(0x1C, outBuffer0.device_address)
    ipFIRN11.write(0x00, 0x01)
    while (ipFIRN11.read(0x00) & 0x4) == 0x0:
        continue
    timeKernelEnd = time()
    print("Kernel execution time: " + str(timeKernelEnd - timeKernelStart) + " s")

    plt.title("FIR Response")
    plt.xlabel("Sample Point")
    plt.ylabel("Magnitude")
    xSeq = range(len(inBuffer0))
    if n32DCGain == 0:
        plt.plot(xSeq, inBuffer0, 'b.', xSeq, outBuffer0, 'r.')
    else:
        plt.plot(xSeq, inBuffer0, 'b.', xSeq, outBuffer0 / n32DCGain, 'r.')
    plt.grid(True)
    plt.show() # In Jupyter, press Tab + Shift keys to show plot then redo run

    print("============================")
    print("Exit process")
```

```
Entry: /usr/local/share/pynq-venv/lib/python3.8/site-packages/ipykernel_launcher.py
System argument(s): 3
Start of "/usr/local/share/pynq-venv/lib/python3.8/site-packages/ipykernel_launcher.py"
Kernel execution time: 0.000274658203125 s
```




```
============================
Exit process
```

#2 :

(1) 內容

#2 和#1 幾乎一樣，只是在讀寫 data 的介面從 AXI-Master 換成 AXI-Stream。接下來只會提到不同的部分。

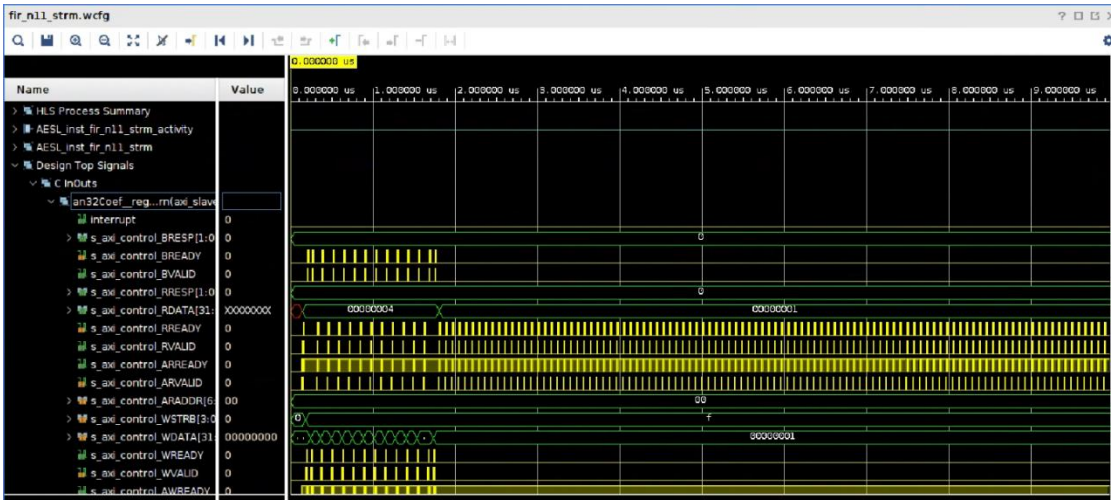
在 vivado 進行 block diagram 的 connection 需要注意的是，要將 dma ip 呼叫出來，並且將其中一個設為 read 另一個設為 write 並提前手動連接 s_ss2m 以及 s_m2ss，將 s_ss2m 連接到 axi_dma_1 而 s_m2ss 連接到 axi_dma_0，接著再讓它進行自動連接，此時一樣要開啟 slave hp port。

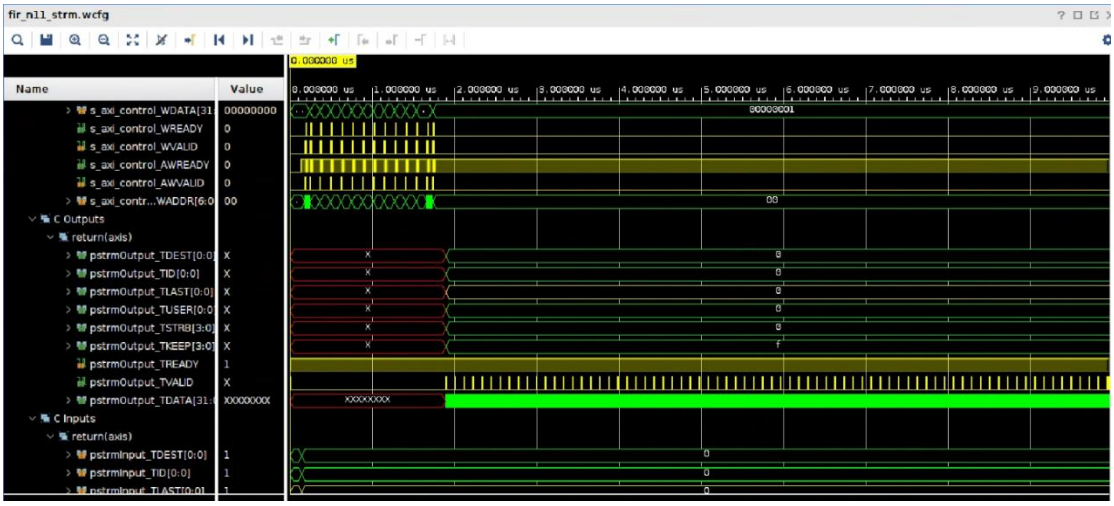
在 python code 的部分，由於其採用 DMA 連結 PS 的 AXI-Master 和 kernel 的 AXI-Stream，因此要透過 ipDMAIn 以及 ipDMAOut 進行 data 讀寫，其他部分和#1 相同。

(2) 相關截圖

```
================================================================
== Synthesis Summary Report of 'fir_nl1_strm'
================================================================
+ General Information:
    * Date:           Fri Feb 28 04:13:58 2025
    * Version:        2022.1 (Build 3526262 on Mon Apr 18 15:47:01 MDT 2022)
    * Project:        hls_FIRN11Stream
    * Solution:       solution1 (Vivado IP Flow Target)
    * Product family: zynquplus
    * Target device:  xck26-sfvc784-2LV-c


+ Performance & Resource Estimates:

    PS: '+' for module; 'o' loop; '*' for dataflow
    +-----------------------------------+-----+-----+--------+--------+---------+--------+-----+----------+-----+---------+-----------+-----------+-----+
    |               Modules             | Issue|      | Latency | Latency| Iteration|        | Trip |          |     |         |           |           |     |
    |               & Loops             | Type | Slack| (cycles)|  (ns)  |  Latency | Interval| Count| Pipelined| BRAM |   DSP   |     FF    |    LUT    | URAM|
    +-----------------------------------+-----+-----+--------+--------+---------+--------+-----+----------+-----+---------+-----------+-----------+-----+
    |+ fir_nl1_strm                     |    -|  1.01|       -|       -|        -|       -|    -|        no|    -|  33 (2%)|  952 (~0%)|  1082 (~0%)|    -|
    | + fir_nl1_strm_Pipeline_XFER_LOOP |    -|  1.01|       -|       -|        -|       -|    -|        no|    -|  33 (2%)|  762 (~0%)|   825 (~0%)|    -|
    |  o XFER_LOOP                      |   II|  7.30|       -|       -|       12|      11|    -|       yes|    -|        -|          -|          -|    -|
    +-----------------------------------+-----+-----+--------+--------+---------+--------+-----+----------+-----+---------+-----------+-----------+-----+


================================================================
== HW Interfaces
================================================================
* S_AXILITE Interfaces
+---------------+------------+---------------+--------+----------+
| Interface     | Data Width | Address Width | Offset | Register |
+---------------+------------+---------------+--------+----------+
| s_axi_control | 32         | 7             | 64     | 0        |
+---------------+------------+---------------+--------+----------+

* S_AXILITE Registers
+---------------+----------------+--------+-------+--------+-----------------------------------+-----------------------------------------------------------------------+
| Interface     | Register       | Offset | Width | Access | Description                       | Bit Fields                                                            |
+---------------+----------------+--------+-------+--------+-----------------------------------+-----------------------------------------------------------------------+
| s_axi_control | CTRL           | 0x00   | 32    | RW     | Control signals                   | 0=AP_START 1=AP_DONE 2=AP_IDLE 3=AP_READY 7=AUTO_RESTART 9=INTERRUPT  |
| s_axi_control | GIER           | 0x04   | 32    | RW     | Global Interrupt Enable Register  | 0=Enable                                                              |
| s_axi_control | IP_IER         | 0x08   | 32    | RW     | IP Interrupt Enable Register      | 0=CHAN0_INT_EN 1=CHAN1_INT_EN                                         |
| s_axi_control | IP_ISR         | 0x0c   | 32    | RW     | IP Interrupt Status Register      | 0=CHAN0_INT_ST 1=CHAN1_INT_ST                                         |
| s_axi_control | regXferLeng    | 0x10   | 32    | W      | Data signal of regXferLeng        |                                                                       |
+---------------+----------------+--------+-------+--------+-----------------------------------+-----------------------------------------------------------------------+

* AXIS
+-------------+--------------+-------+-------+----+----+-------+-------+-------+-------+-------+
| Interface   | Register Mode | TDATA | TDEST | TID | TKEEP | TLAST | TREADY | TSTRB | TUSER | TVALID |
+-------------+--------------+-------+-------+----+-------+-------+-------+-------+-------+-------+
| pstrmInput  | both         | 32    | 1     | 1  | 4     | 1     | 1     | 4     | 1     | 1     |
| pstrmOutput | both         | 32    | 1     | 1  | 4     | 1     | 1     | 4     | 1     | 1     |
+-------------+--------------+-------+-------+----+-------+-------+-------+-------+-------+-------+

* TOP LEVEL CONTROL
+-----------+------------+-----------+
| Interface | Type       | Ports     |
+-----------+------------+-----------+
| ap_clk    | clock      | ap_clk    |
| ap_rst_n  | reset      | ap_rst_n  |
| interrupt | interrupt  | interrupt |
| ap_ctrl   | ap_ctrl_hs |           |
+-----------+------------+-----------+
```

```
================================================================
== Vitis HLS Report for 'fir_nl1_strm'
================================================================
* Date:           Fri Feb 28 04:13:58 2025

* Version:        2022.1 (Build 3526262 on Mon Apr 18 15:47:01 MDT 2022)
* Project:        hls_FIRN11Stream
* Solution:       solution1 (Vivado IP Flow Target)
* Product family: zynquplus
* Target device:  xck26-sfvc784-2LV-c


================================================================
== Performance Estimates
================================================================
+ Timing:
    * Summary:
    +--------+---------+----------+------------+
    |  Clock |  Target | Estimated| Uncertainty|
    +--------+---------+----------+------------+
    |ap_clk  |  10.00 ns|  6.290 ns|     2.70 ns|
    +--------+---------+----------+------------+

+ Latency:
    * Summary:
    +---------+---------+----------+----------+-----+-----+---------+
    |  Latency (cycles) |  Latency (absolute) |  Interval | Pipeline|
    |   min   |   max   |    min   |    max   | min | max |   Type  |
    +---------+---------+----------+----------+-----+-----+---------+
    |        ?|        ?|         ?|         ?|    ?|    ?|       no|
    +---------+---------+----------+----------+-----+-----+---------+

    + Detail:
        * Instance:
        +-----------------------------------------+-----------------------------------+---------+---------+---------+---------+-----+-----+---------+
        |                                         |                                   | Latency (cycles) | Latency (absolute) |  Interval | Pipeline|
        |                Instance                 |              Module               |   min   |   max   |    min  |    max  | min | max |   Type  |
        +-----------------------------------------+-----------------------------------+---------+---------+---------+---------+-----+-----+---------+
        |grp_fir_nl1_strm_Pipeline_XFER_LOOP_fu_112 |fir_nl1_strm_Pipeline_XFER_LOOP |        ?|        ?|        ?|        ?|    ?|    ?|       no|
        +-----------------------------------------+-----------------------------------+---------+---------+---------+---------+-----+-----+---------+

        * Loop:
        N/A


================================================================
== Utilization Estimates
================================================================
* Summary:
+---------------+---------+-----+-------+-------+-----+
|      Name     | BRAM_18K|  DSP|   FF  |   LUT | URAM|
+---------------+---------+-----+-------+-------+-----+
|DSP            |        -|    -|      -|      -|    -|
|Expression     |        -|    -|      0|     42|    -|
|FIFO           |        -|    -|      -|      -|    -|
|Instance       |        0|   33|    916|   1005|    0|
|Memory         |        -|    -|      -|      -|    -|
|Multiplexer    |        -|    -|      -|     35|    -|
|Register       |        -|    -|     36|      -|    -|
+---------------+---------+-----+-------+-------+-----+
|Total          |        0|   33|    952|   1082|    0|
+---------------+---------+-----+-------+-------+-----+
|Available      |      288| 1248| 234240| 117120|  64|
+---------------+---------+-----+-------+-------+-----+
|Utilization (%)|        0|    2|     ~0|     ~0|    0|
+---------------+---------+-----+-------+-------+-----+
```



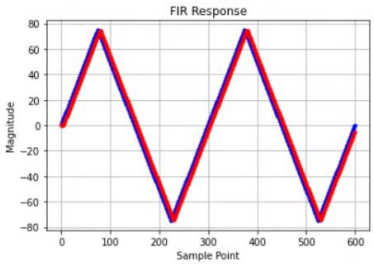


In [6]:

```python
# coding: utf-8

# In[ ]:


from __future__ import print_function

import sys, os
import numpy as np
from time import time
import matplotlib.pyplot as plt

sys.path.append('/home/xilinx')
os.environ['XILINX_XRT'] = '/usr'
from pynq import Overlay
from pynq import allocate

if __name__ == "__main__":
    print("Entry:", sys.argv[0])
    print("System argument(s):", len(sys.argv))

    print("Start of \"" + sys.argv[0] + "\"")

    ol = Overlay("/home/root/jupyter_notebooks/FIRN11MAXI.bit")
    ipFIRN11 = ol.fir_n11_maxi_0

    fiSamples = open("samples_triangular_wave.txt", "r+")
    numSamples = 0
    line = fiSamples.readline()
    while line:
        numSamples = numSamples + 1
        line = fiSamples.readline()

    inBuffer0 = allocate(shape=(numSamples,), dtype=np.int32)
    outBuffer0 = allocate(shape=(numSamples,), dtype=np.int32)
    fiSamples.seek(0)
    for i in range(numSamples):
        line = fiSamples.readline()
        inBuffer0[i] = int(line)
    fiSamples.close()

    numTaps = 11
    n32Taps = [0, -10, -9, 23, 56, 63, 56, 23, -9, -10, 0]
    #n32Taps = [1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1]
    n32DCGain = 0
    timeKernelStart = time()
    for i in range(numTaps):
        n32DCGain = n32DCGain + n32Taps[i]
        ipFIRN11.write(0x40 + i * 4, n32Taps[i])
    if n32DCGain < 0:
```

```python
    if n32DCGain < 0:
        n32DCGain = 0 - n32DCGain
    ipFIRN11.write(0x28, len(inBuffer0) * 4)
    ipFIRN11.write(0x10, inBuffer0.device_address)
    ipFIRN11.write(0x1C, outBuffer0.device_address)
    ipFIRN11.write(0x00, 0x01)
    while (ipFIRN11.read(0x00) & 0x4) == 0x0:
        continue
    timeKernelEnd = time()
    print("Kernel execution time: " + str(timeKernelEnd - timeKernelStart) + " s")

    plt.title("FIR Response")
    plt.xlabel("Sample Point")
    plt.ylabel("Magnitude")
    xSeq = range(len(inBuffer0))
    if n32DCGain == 0:
        plt.plot(xSeq, inBuffer0, 'b.', xSeq, outBuffer0, 'r.')
    else:
        plt.plot(xSeq, inBuffer0, 'b.', xSeq, outBuffer0 / n32DCGain, 'r.')
    plt.grid(True)
    plt.show() # In Jupyter, press Tab + Shift keys to show plot then redo run

    print("============================")
    print("Exit process")
```

```
Entry: /usr/local/share/pynq-venv/lib/python3.8/site-packages/ipykernel_launcher.py
System argument(s): 3
Start of "/usr/local/share/pynq-venv/lib/python3.8/site-packages/ipykernel_launcher.py"
Kernel execution time: 0.000274658203125 s
```




```
============================
Exit process
```